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Information

Bayesian Classification

Consider minimum risk classification as introduced in the lecture, with the Conditional Risk formula reproduced below, for your convenience:

$$R(\alpha_i | \mathbf{x}) = \sum_{j=1}^c \lambda(\alpha_i | \omega_j) P(\omega_j | \mathbf{x})$$

Assume we have $\Omega = \{\omega_1, \dots, \omega_c\}$, a list of actions $\{\alpha_i = \text{"predict class } \omega_i \text{"}\}$, a given cost function $\lambda = \{\lambda_{ij} = \lambda(\alpha_i | \omega_j)\}$, and known probability distributions $P(\Omega)$ and $P(\mathbf{X} | \Omega)$.

Information

Which of the following statements are correct?

Question **1**

Correct

Mark 1.00 out of 1.00

MAP classification behaviour can be obtained by an appropriate setting of the cost function

Select one:

☒ True ✓

☐ False

Question **2**

Correct

Mark 1.00 out of 1.00

The effect of any change to the cost function can also be achieved by leaving the cost function unchanged and modifying the likelihoods $P(\mathbf{X} | \omega_j)$ accordingly

Select one:

☐ True

☒ False ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Written as a matrix, the above cost function λ is of quadratic shape

Select one:

☒ True ✓☐ FalseQuestion **4**

Correct

Mark 1.00 out of 1.00

For any cost function, the expected cost is equal to the Bayes Error

Select one:

☐ True☒ False ✓Question **5**

Correct

Mark 1.00 out of 1.00

The expected risk defines decision boundaries only in a two-dimensional feature space

Select one:

☐ True☒ False ✓

Information

Answer the following questions:

Question **6**

Incorrect

Mark 0.00 out of 1.00

If we multiply the cost function with a constant k , the probability of all classes will
(multiple answers possible)

- ☐ remain the same
- ☐ sum to $1.0 * k$
- ☒ change so that probable classes become less probable, and vice versa ("smoothing") ✖
- ☐ increase by a factor k
- ☒ be reduced by a factor k ✖

Your answer is incorrect.

Question **7**

Incorrect

Mark 0.00 out of 1.00

If we set the cost function to all 1 (i.e., all $\lambda_{ij} = 1$),
(multiple answers possible)

- ☐ minimum risk classification becomes the same as MAP classification
- ☒ the risk for all actions becomes c , for any $\mathbf{x}$, where $c > 1$ is the number of classes ✖
- ☐ all decisions will have the same cost, for any $\mathbf{x}$
- ☐ the risk for all actions becomes 1, for any $\mathbf{x}$
- ☐ for any $\mathbf{x}$, every decision is equally good

Your answer is incorrect.

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Classifier Evaluation

Consider a two-class classification problem with classes p and n , and a test set T with 1000 test examples and class distribution $P:N = 3:7$. For each of the following classifiers, specify what the expected confusion matrix will look like when the classifier is tested on T (give the numbers for the four entries of the table; arithmetic expressions (e.g. $3/5$ instead of 0.6) are also accepted). For each classifier, also give the expected values for **Recall** and **Precision**. If one of these is **not defined**, enter -1.

Question 8

Not answered

Marked out of 3.00

A classifier that randomly predicts n or p with probability 0.5

TP:  FP: FN:  TN: Recall: Precision: **Question 9**

Not answered

Marked out of 3.00

A classifier that always predicts class n

TP:  FP: FN:  TN: Recall: Precision: 

Information

Which of the following statements are true:

Question **10**

Correct

Mark 1.00 out of 1.00

The iso-performance line cannot go through points $(0, 0)$ and $(1, 1)$

Select one:

- ☐ True
- ☒ False ✓

Question **11**

Correct

Mark 1.00 out of 1.00

For any classifier C and any dataset $\mathcal{D}$, $Recall(C, \mathcal{D}) + Precision(C, \mathcal{D}) = 1.0$

Select one:

- ☐ True
- ☒ False ✓

Question **12**

Correct

Mark 1.00 out of 1.00

If two classifiers C_1, C_2 have the same True Positive Rates (TPRs), but different False Positive Rates, they cannot have the same accuracy on any given fixed test set T .

Select one:

- ☒ True ✓
- ☐ False

Question **13**

Correct

Mark 1.00 out of 1.00

Two classifiers with the same accuracy cannot occupy different places in ROC space.

Select one:

- ☐ True
- ☒ False ✓

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Decision Trees

Consider the following alternative criterion (instead of information gain) for evaluating a feature A for splitting in the ID3 algorithm, in a binary classification task with two classes $\Omega = \{+, -\}$ and $p_i(n_i)$ = number of pos (neg) examples in $\mathcal{D}_i$, the i -th branch (notation as in the lecture slides):

$$G_{\mathcal{D}}(A) = \sum_{v_i \in A} \frac{|\mathcal{D}_i|}{|\mathcal{D}|} \cdot \frac{|p_i - n_i|}{p_i + n_i}$$

Answer the following questions:

Question 14

Incorrect

Mark 0.00 out of 1.00

The best feature is the one that ✖ this measure

Your answer is incorrect.

Question 15

Incorrect

Mark 0.00 out of 1.00

This measure will become maximal when each branch only contains examples of one class

Select one:

- ☐ True
- ☒ False ✖

Question 16

Incorrect

Mark 0.00 out of 1.00

When a feature A perfectly discriminates between the classes in $\mathcal{D}$, $G_{\mathcal{D}}(A)$

- ☐ < 0.0
- ☐ is useless
- ☐ $= 1.0$
- ☐ is undefined
- ☐ is equal to the information gain
- ☐ > 1.0
- ☒ $= 0.0$ ✖

Your answer is incorrect.

Information

Consider a variant "ID3-S" of the ID3 algorithm that learns only trees that are at most one level deep. It chooses a feature for the root, using the Information Gain heuristic, and labels each of the resulting nodes with the class that is most frequent in that node (in case of a tie, it randomly chooses between one of the competing classes).

Which of the following statements about ID3-S are true:

Question 17

Correct

Mark 1.00 out of 1.00

This is not a valid learning algorithm

Select one:

- ☐ True
- ☒ False ✔

Question 18

Correct

Mark 1.00 out of 1.00

This algorithm cannot result in a classifier with an accuracy of 1.0 on the training set

Select one:

- ☐ True
- ☒ False ✔

Question **19**

Correct

Mark 1.00 out of 1.00

This is a high-bias learner

Select one:

☒ True ✓☐ FalseQuestion **20**

Correct

Mark 1.00 out of 1.00

Combining different such trees via voting can improve accuracy on some dataset

Select one:

☒ True ✓☐ False[◀ Virtual lecture hall for the exam](#)[First Exam ▶](#)

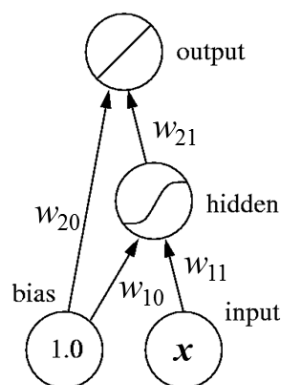
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Neural Networks and Deep Learning

Consider the simple neural network from the lecture slides, with 1 input unit (for a feature x), one hidden unit, and one output unit y , where the hidden unit uses the $\tanh$ as activation function, and the output unit y uses the identity function (i.e., just outputs what comes in as net input).

Also assume that the loss function is the sum squared error (as defined in the slides).



Which of the following statements are correct for this network?

Question 21

Correct

Mark 1.00 out of 1.00

A training set for this network will have to consist of pairs of numbers

Select one:

- ☒ True ✓
☐ False

Question 22

Incorrect

Mark 0.00 out of 1.00

Changing the output activation function from linear (identity) to $\tanh$ will not make any difference, as the hidden unit already uses the $\tanh$ activation

Select one:

- ☒ True ✗
☐ False

Question 23

Incorrect

Mark 0.00 out of 1.00

If we change the output of the bias unit from 1.0 to 2.0, the result is no longer a valid neural network

Select one:

- ☒ True ✖
- ☐ False

Question 24

Correct

Mark 1.00 out of 1.00

Changing the output of the bias unit from 1.0 to 2.0 will change the absolute value of the Sum Squared Error on the training set $\mathcal{D}$, but will not change the values of the network weights that minimise the error on $\mathcal{D}$

Select one:

- ☐ True
- ☒ False ✔

Question 25

Incorrect

Mark 0.00 out of 1.00

The loss function is defined over a four-dimensional parameter space

Select one:

- ☐ True
- ☒ False ✖

Information

Now assume we use the above simple network for a binary classification task. We encode the class labels in the training data $\mathcal{D}$ as 1 and 0, respectively, equip the output unit y with a logistic activation function (see slides), and use the **Binary Cross-Entropy CE** (as defined in the lecture slides) as the loss function:

$$CE(\mathbf{x}_i) = -y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i)$$

where y_i is the true class of example $\mathbf{x}_i$ (0 or 1), and $\hat{y}_i$ is the value predicted by output unit y .

An example $\mathbf{x}_i$ is **classified** as class 1 iff the output $\hat{y}_i$ calculated by the network from input x is ≥ 0.5 .

Which of the following statements are correct?

Question **26**

Correct

Mark 1.00 out of 1.00

This classifier can only learn decision boundaries that split the feature space into two regions

Select one:

- ☒ True ✓
- ☐ False

Question **27**

Correct

Mark 1.00 out of 1.00

If we make the network deeper by adding additional hidden layers, the CE function must be extended with corresponding additional terms

Select one:

- ☐ True
- ☒ False ✓

Question **28**

Incorrect

Mark 0.00 out of 1.00

If the class distribution in the training set is 50:50, the binary cross-entropy CE encourages the network to predict values close to 0.5

Select one:

- ☒ True ✗
- ☐ False

Question **29**

Incorrect

Mark 0.00 out of 1.00

This classifier can easily overfit because the cross-entropy is non-linear

Select one:

- ☒ True ✗
- ☐ False

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Unsupervised Learning

Consider the task of k-means clustering with N observations described in terms of D numeric features, with the Euclidean distance as distance measure. Assume we have set the number k of clusters to $k = 5$.

Which of the following statements are correct?

Question **30**

Incorrect

Mark 0.00 out of 1.00

When we increase the number of features, the model complexity penalty term in the Bayes Information Criterion will automatically grow larger

Select one:

- ☐ True
- ☒ False 

Question **31**

Correct

Mark 1.00 out of 1.00

In k-means, k has to be at least $N/2$

Select one:

- ☐ True
- ☒ False 

Question **32**

Correct

Mark 1.00 out of 1.00

To use the Bayes Information Criterion for selecting an appropriate value for k , a k-means cluster model must be converted into a probabilistic model

Select one:

- ☒ True 
- ☐ False

Question 33

Correct

Mark 1.00 out of 1.00

The cluster with the smallest number of members will have the lowest (best) MQE

Select one:

- ☐ True
- ☒ False ✓

Question 34

Correct

Mark 1.00 out of 1.00

Feature normalisation will change the MQE, but not the clusters found

Select one:

- ☐ True
- ☒ False ✓

Information

Consider a Gaussian Mixture Model (GMM) with k components that was learned from a dataset of N examples over a D -dimensional numeric feature space.

Question 35

Incorrect

Mark 0.00 out of 1.00

How many parameters (numbers) does a specification of this GMM consist of (ignoring possible redundancies because of distribution constraints and symmetries)?

- ☐ $k + kD + D$
- ☐ $k(D \cdot N)$
- ☐ $k + kD + kD^2$
- ☐ $3kD$
- ☒ $kD + N$ ✗
- ☐ $kN + kD + kD^2$
- ☐ $3k$

Your answer is incorrect.

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Markov Models

Consider a simple Markov process with 3 states (S1, S2, S3), the following state transition model $\mathbf{A}$, and a uniform prior state distribution $\boldsymbol{\pi} = [1/3, 1/3, 1/3]$:

$$\mathbf{A} = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.1 & 0.8 \end{bmatrix}$$

Which of the following statements are true?

Question 36

Incorrect

Mark 0.00 out of 1.00

If we changed some a_{ij} to be zero (0.0), then also a_{ji} would have to be changed to zero for a valid Markov process

Select one:

- ☒ True ✖
- ☐ False

Question 37

Incorrect

Mark 0.00 out of 1.00

If we made one of these numbers smaller, then at least one number in the same column would have to become larger

Select one:

- ☒ True ✖
- ☐ False

Question 38

Incorrect

Mark 0.00 out of 1.00

If we made one of these numbers smaller, then at least one number in the same row would have to become larger

Select one:

- ☐ True
- ☒ False ✖



Question 39

Correct

Mark 1.00 out of 1.00

The expected number of consecutive occurrences of state S2 is higher than the expected number of consecutive occurrences of state S1

Select one:

☒ True ✓

☐ False

Information

Now consider a simple Markov process with 3 states (S1, S2, S3), the following state transition model $\mathbf{A}$, and a uniform prior state distribution $\boldsymbol{\pi} = [1/3, 1/3, 1/3]$:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

What is the probability of the following state sequences, according to this model M :

Question 40

Incorrect

Mark 0.00 out of 1.00

$$P((S1, S1, S2, S2, S3, S3) \mid M) = \dots$$

Answer: 0.5



Question 41

Incorrect

Mark 0.00 out of 1.00

$$P((S1, S1, S1, S1, S1) \mid M) = \dots$$

Answer: 0.5



Question 42

Incorrect

Mark 0.00 out of 1.00

$$P((S1, S2, S3, S1, S2, S3) \mid M) = \dots$$

Answer: 0.5



Information

Consider the classification task of identifying one of two cities A or B, based on a sequence of weather observations. We have learned two Markov models M_A and M_B that model the weather in cities A and B, respectively. Now, given a sequence S of 10 consecutive weather observations, you want to decide which of the two cities this sequence is from, based on the MAP principle. Model M_A stands for city A, M_B for B.

Question 43

Partially correct

Mark 0.50 out of 1.00

Which of the following quantities are needed for making this decision?

- ☐ $P(A)$
- ☒ $P(S|M_B)$ ✓
- ☐ $P(M_B|M_A)$
- ☒ $P(S|M_A)$ ✓
- ☐ $P(B)$

Your answer is partially correct.

You have correctly selected 2.

Information

Which of the following statements are true?

Question 44

Correct

Mark 1.00 out of 1.00

If we produce a new sequence S_2 by concatenating two copies of S , the likelihoods will change, but the MAP decision will always remain the same

Select one:

- ☐ True
- ☒ False ✓



Question **45**

Correct

Mark 1.00 out of 1.00

This kind of classification procedure is only possible if the length of S is at least 2

Select one:

- ☐ True
- ☒ False ✓

Question **46**

Incorrect

Mark 0.00 out of 1.00

The same decision could be made without a Markov model, by counting the different states in S

Select one:

- ☒ True ✗
- ☐ False

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